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Earth Science

EARTH SCIENCE FIELDTRIP REPORT

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Contents

1 Introduction **4**
- 1.1 Purpose of Preparing the Field Trip Report . . . 4
- 1.2 Task Assignment Table of Group Members . . . 4
- 1.3 Natural, Technical, and Socio-economic Characteristics of the Study Area . . . 4
- 1.4 Tentative Schedule . . . 5
  - 1.4.1 General Planning Schedule . . . 5
  - 1.4.2 Actual Fieldtrip Activities . . . 5

2 Geosphere **5**
- 2.1 Weathering in Laterite and Formation Mechanism . . . 5
- 2.2 Erosion and Coastal Processes . . . 7
- 2.3 Magma Eruption Processes and Rhyolite . . . 7
- 2.4 Granitic Intrusion, Diabase Dykes, and Cross-cutting Relationships . . . 8
  - 2.4.1 Granitic Intrusion . . . 8
  - 2.4.2 Diabase Dykes . . . 8
  - 2.4.3 Cross-cutting Relationship of Intrusive Igneous Rocks . . . 8
- 2.5 Summary of Geosphere Observations . . . 9

3 Hydrosphere **9**
- 3.1 Surface Water and Stream Processes at Suoi Tien (Fairy Stream) . . . 9
- 3.2 Groundwater Dynamics in Coastal Sand Dune Systems (Ho Tram & Coastal Dunes) . . . 10
- 3.3 Coastal Erosion, Wave Action, and Marine Processes (Back Beach, Mui Ke Ga, & Ho Tram Estuary) . . . 11
- 3.4 Impact of Mining Pit Water Retention at Ho Da and Gia Quy Quarry . . . 11

4 Biosphere **13**
- 4.1 Biological Weathering by Tree Roots . . . 13
- 4.2 Mangrove Forests and Coastal Protection . . . 13

5 Atmosphere **14**
- 5.1 Description of Typhoon Characteristics in Vietnam . . . 14
  - 5.1.1 Number of Storms per Year . . . 14
  - 5.1.2 Time of Typhoon Occurrence . . . 14
  - 5.1.3 Magnitude and Intensity of Typhoons . . . 14
  - 5.1.4 Affected Areas . . . 14
  - 5.1.5 Frequency of Strong Typhoons . . . 15
  - 5.1.6 Consequences of Typhoons . . . 15
- 5.2 Rainfall Characteristics in Vietnam by Region . . . 15
  - 5.2.1 Rainfall in Northern Vietnam . . . 15
  - 5.2.2 Rainfall in Central Vietnam . . . 15
  - 5.2.3 Rainfall in Southern Vietnam . . . 16
  - 5.2.4 Comparison Among Regions . . . 16
- 5.3 Impacts of Storms and Rainfall on Landslides and Flooding Hazards . . . 16
  - 5.3.1 Landslides . . . 16
  - 5.3.2 Flooding . . . 16
  - 5.3.3 Relationship with Field Observations . . . 17

6 Natural Resources **17**
- 6.1 VNU-HCM Quarry and Environmental Restoration Efforts . . . 17
- 6.2 Basalt and Pozzolan Resources at Gia Quy Quarry . . . 17
- 6.3 Petroleum Service Industry in Vung Tau: Vietsovpetro Joint Venture . . . 18

7 Conclusion **19**

List of Figures

1 Official fieldtrip schedule for Earth Science (GE1013). 5
2 Laterite sample showing reddish-brown color and rough weathered texture. 7
3 Wind-shaped sand surface and ripple-like structures observed at Cua Lap. 7
4 Rhyolite surface observed near the coastal area of Vung Tau. 8
5 Fractured and weathered rhyolite outcrop near the coastal area of Vung Tau. 8
6 Light-colored granite observed at Sao Mai Quarry. 8
7 Dark-colored diabase observed at Sao Mai Quarry. 9
8 Dark diabase cutting through light-colored granite at Sao Mai Quarry. 9
9 Surface water flow and undercut sand cliffs observed at Suoi Tien. 10
10 Aeolian sand dune surface acting as a groundwater recharge zone at Ho Tram. 10
11 Eroded sandy beach scarp and wave impact observed along the coast. 11
12 Wave action against rocky granite headlands observed at Mui Ke Ga. 12
13 Accumulated surface and groundwater filling the former quarry pit at Ho Da. 12
14 Plant growth along rock fractures, demonstrating biological weathering caused by roots. . 13
15 Mangrove forest along a water channel, showing dense vegetation in a coastal wetland environment. 13
16 Mangrove vegetation near a river-mouth environment, illustrating its role in stabilizing sediments and protecting coastal areas. 14
17 VNU-HCM Quarry, also known as Ho Da, showing the abandoned quarry lake and surrounding rock slopes. 17
18 Exposed rock slope and vegetation around the former VNU-HCM Quarry. 17
19 Basaltic materials and quarry activity observed at Gia Quy Quarry. 18
20 Basalt samples collected at Gia Quy Quarry, showing dark color and vesicular texture. . 18
21 Pozzolan material observed at Gia Quy Quarry, which can be used as an additive in cement production. 18
22 Model of offshore oil fields and petroleum infrastructure displayed at Vietsovpetro. . . . 19
23 Offshore petroleum platform model displayed at Vietsovpetro. 19
24 Geological samples and petroleum-related materials displayed at Vietsovpetro. 19
25 Scale model of the Vietsovpetro area and facilities. 20

List of Tables

1 Task assignment of group members 4
2 Overview of actual fieldtrip activities and geological topics 6

1 Introduction

1.1 Purpose of Preparing the Field Trip Report

The purpose of this field trip report is to summarize, organize, and interpret the observations collected during the Earth Science fieldtrip. Instead of describing the visited locations only in the order of the itinerary, this report follows the structure of Earth systems, including the geosphere, hydrosphere, biosphere, atmosphere, and natural resources.

Through the fieldtrip, students had the opportunity to observe geological materials and natural processes directly in real environments. The report focuses on connecting field evidence with theoretical knowledge learned in class, such as rock formation, weathering, sediment transportation, coastal processes, groundwater, and the use of geological resources in human activities.

Preparing this report also helped students develop important academic and fieldwork skills. These include taking field notes, collecting rock and sediment samples, using maps and compasses, taking photographs, recording videos, and interpreting geological features based on real observations. Therefore, the report serves not only as a summary of the fieldtrip but also as a reflection of the learning process and teamwork during the course.

1.2 Task Assignment Table of Group Members

During the fieldtrip, each member was assigned specific responsibilities to make sure that observations were recorded systematically and samples were collected properly. Although each student had a main task, all members supported one another during observation, discussion, sample collection, and documentation.

Table 1: Task assignment of group members

No.	Name	Main Tasks
1	Nguyễn Trung Nhân	Following the lecturer closely and collecting rock samples.
2	Nguyễn Vĩnh Phú	Collecting samples and extracting sample information for observation.
3	Nguyễn Kim Hoàng Phi	Collecting samples during field observations.
4	Phan Nguyễn Khánh Mai	Taking notes and taking photos during the fieldtrip.
5	Nguyễn Thị Yến Nhi	Taking notes and recording videos during the fieldtrip.

1.3 Natural, Technical, and Socio-economic Characteristics of the Study Area

The fieldtrip was conducted in Vung Tau and nearby areas, where students could observe a combination of natural landscapes, geological outcrops, coastal environments, and industrial facilities. The study area includes old and active quarries, coastal beaches, sand dune systems, hillside terrain, and petroleum-related industrial sites. These locations provided suitable conditions for studying Earth Science concepts in real field settings.

From a natural perspective, the study area contains diverse geological materials and landforms. At VNU Lake, students observed andesite, an extrusive igneous rock related to volcanic activity. At Gia Quy Quarry, basaltic materials, pozzolan, and spheroidal weathering features were observed. At Sao Mai Quarry, granite and diabase provided evidence of intrusive igneous activity and cross-cutting relationships. Coastal locations such as Back Beach and Cua Lap showed marine sediments, aeolian sand dunes, coastal transportation, and deposition processes. The Lighthouse area provided a broader view of hillside terrain and coastal geomorphology.

From a technical perspective, the fieldtrip helped students practice basic geological field methods. These included collecting rock and sediment samples, using maps and compasses, recording field notes,

taking photographs, and observing rock structures and sedimentary features. The use of smartphones and the Clino app also supported basic field measurements and documentation.

From a socio-economic perspective, the study area demonstrates the relationship between geology and human activities. Gia Quy Quarry shows the extraction and use of basaltic and pozzolanic materials for construction and cement-related applications. Vietsovpetro represents the petroleum service industry in Vung Tau and shows how geological knowledge is applied in offshore exploration, energy production, and industrial development. Therefore, the fieldtrip area is valuable not only for geological study but also for understanding the practical role of Earth Science in society.

1.4 Tentative Schedule

The Earth Science fieldtrip was organized over three days in Vung Tau and nearby coastal areas. Before the departure, a general fieldtrip plan was prepared to ensure that all activities, observation sites, and learning objectives could be completed effectively. The schedule included geological observations at igneous rock outcrops, coastal sedimentary environments, mining areas, and industrial locations related to petroleum geology.

The fieldtrip was designed to help students connect theoretical geological knowledge with practical field observations. At each location, students conducted geological observations, identified rock and sediment characteristics, analyzed geological structures, and discussed geological processes under the guidance of the lecturers.

1.4.1 General Planning Schedule

The following figure presents the official fieldtrip schedule prepared before the excursion.

[Image: Official fieldtrip schedule for Earth Science (GE1013)]

Figure 1: Official fieldtrip schedule for Earth Science (GE1013).

The planned schedule included several important observation sites such as VNU Lake, Gia Quy Quarry, Sao Mai Quarry, Back Beach, Cua Lap, Vietsovpetro, and the Lighthouse area. Each site focused on different geological topics including igneous rocks, weathering, sedimentary processes, coastal geology, structural relationships, and industrial applications of geology.

1.4.2 Actual Fieldtrip Activities

During the fieldtrip, students participated in geological observations and practical activities at each scheduled stop. The activities included rock identification, sediment observation, structural interpretation, sample collection, field documentation, and discussions with lecturers regarding geological processes and environmental conditions.

Overall, the fieldtrip activities were completed successfully and provided students with valuable practical experience in geological observation and interpretation. The combination of field investigation, group discussion, and direct interaction with geological features helped strengthen students' understanding of Earth Science concepts learned in class.

2 Geosphere

2.1 Weathering in Laterite and Formation Mechanism

Laterite is a weathering product commonly formed in tropical regions where the climate is hot and humid. Under these conditions, rainwater infiltrates into the ground and causes strong chemical weathering of the original parent rock. During this process, soluble elements such as silica, calcium, sodium, and

Table 2: Overview of actual fieldtrip activities and geological topics

Date/Time	**Location**	**Coordinates**	**Main Activities**	**Main Topics**
7:00 AM May 8, 2026	Ho Da VNU	Latitude: 10.877268°N Longitude: 106.790556°E	Observing andesite, describing rock characteristics, and practicing field observation techniques.	Andesite; laterite
11:00 AM May 8, 2026	Gia Quy Quarry	Latitude: 10.503907°N Longitude: 107.272151°E	Identifying basalt, observing volcanic textures, weathering features, and the mining environment.	Basalt; weathering; mining
2:30 PM May 8, 2026	Sao Mai Quarry	Latitude: 10.385507°N Longitude: 107.063481°E	Observing granite and diabase, studying cross-cutting relationships, and collecting rock samples.	Granite; diabase; intrusion
5:30 PM May 8, 2026	Back Beach	Latitude: 10.328259°N Longitude: 107.088303°E	Observing marine sediments, analyzing beach sand, and studying coastal transportation and deposition.	Marine sediments; coastal processes
7:00 AM May 9, 2026	Cua Lap	Latitude: 10.390890°N Longitude: 107.158205°E	Studying coastal sediments, Aeolian sand dunes, groundwater conditions, and sedimentary processes.	Aeolian sediments; groundwater
9:30 AM May 9, 2026	Vietsovpetro	Latitude: 10.360200°N Longitude: 107.084696°E	Attending an industry visit, learning about petroleum-related applications of geology, and joining an orientation session.	Petroleum industry; energy resources
7:00 AM May 10, 2026	Lighthouse	Latitude: 10.334079°N Longitude: 107.077631°E	Observing coastal landscapes, hillside terrain, and panoramic views of Vung Tau City and the surrounding sea during the Lighthouse observation activity.	Coastal geomorphology; landscape

potassium are gradually leached away, while less soluble iron and aluminum compounds remain and become concentrated.

The reddish-brown to yellowish color of laterite is mainly caused by the enrichment of iron oxides and hydroxides. These minerals form when iron-bearing minerals in the parent rock react with oxygen and water. As weathering continues, the material becomes harder and more compact, producing a rough and nodular surface. This explains why laterite often appears as a reddish, porous, and strongly weathered rock-like material.

In this fieldtrip, laterite was observed as a weathered material with brown, yellow, and reddish tones. The sample showed a rough surface with many small particles and iron-rich patches, indicating long-term chemical weathering under tropical environmental conditions.

[Image: Laterite sample]

Figure 2: Laterite sample showing reddish-brown color and rough weathered texture.

The observation of laterite helped students understand that weathering is not only a process of breaking rocks into smaller pieces, but also a chemical transformation of minerals. In this case, the strong leaching process removed mobile elements and left behind iron-rich materials, forming laterite.

2.2 Erosion and Coastal Processes

Coastal erosion is the process in which waves, currents, tides, and wind remove and transport sediments or rock materials from the shoreline. In coastal environments, erosion and deposition often occur together. While some areas lose sediments due to wave attack, other areas receive and accumulate transported materials.

At Cua Lap, students observed sandy surfaces shaped by wind and coastal processes. The sand surface showed ripple-like patterns, which are small sedimentary structures formed by the movement of wind over loose sand. These features indicate that wind is an important transportation agent in this area. Dry sand particles are moved across the surface and gradually accumulate to form aeolian sand deposits.

[Image: Wind-shaped sand surface]

Figure 3: Wind-shaped sand surface and ripple-like structures observed at Cua Lap.

The observation at Cua Lap demonstrates the interaction between the atmosphere, hydrosphere, and geosphere. Wind transports dry sand inland, while waves and coastal currents continuously rework beach sediments near the shoreline. As a result, the coastal area is constantly modified by erosion, transportation, and deposition.

These processes are important because they control the shape of beaches, sand dunes, and coastal landforms. They also influence coastal stability and may affect human activities in nearby areas.

2.3 Magma Eruption Processes and Rhyolite

Rhyolite is an extrusive igneous rock formed from silica-rich lava. It is produced when magma erupts onto or near the Earth's surface and cools rapidly. Because rhyolitic magma contains a high amount of silica, it is usually more viscous than basaltic magma. As a result, rhyolite is commonly associated with volcanic activity, lava domes, and sometimes explosive eruptions.

During the fieldtrip, rhyolite was observed in the coastal area of Vung Tau. The rock surfaces showed dark grey, purplish, reddish-brown, and weathered colors. These color variations may be related to mineral composition, oxidation of iron-bearing minerals, and long-term exposure to coastal weathering conditions. The observed rhyolite also showed fractures and irregular rock blocks, indicating that the rock had been affected by both volcanic formation processes and later surface weathering.

The rhyolite outcrop near the coast provided a clear example of volcanic rock exposed at the Earth's surface. Since rhyolite forms from erupted lava, it represents past magma eruption processes in the

[Image: Rhyolite surface]

Figure 4: Rhyolite surface observed near the coastal area of Vung Tau.

geological history of the area. Compared with basalt, rhyolite is generally richer in silica and may display lighter, reddish, or purplish tones depending on oxidation and alteration.

In addition, the coastal setting influenced the appearance of the rhyolite outcrop. Sea waves, salt-water, wind, rainfall, and temperature changes can gradually weather exposed rock surfaces and widen existing cracks. Therefore, the site shows how an igneous rock formed by endogenous processes can later be modified by exogenous processes such as weathering and coastal erosion.

[Image: Fractured and weathered rhyolite]

Figure 5: Fractured and weathered rhyolite outcrop near the coastal area of Vung Tau.

The observation helped students understand that rhyolite is not only evidence of past volcanic activity, but also a rock type that continues to be changed after formation. The fractures, weathered surfaces, and coastal exposure demonstrate the combined effects of magma eruption, structural cracking, oxidation, and coastal weathering.

2.4 Granitic Intrusion, Diabase Dykes, and Cross-cutting Relationships

2.4.1 Granitic Intrusion

Granite is an intrusive igneous rock formed when magma cools slowly beneath the Earth's surface. Because the cooling process happens underground and takes a long time, mineral crystals have enough time to grow. Therefore, granite usually has a coarse-grained texture, with visible mineral grains such as quartz and feldspar.

At Sao Mai Quarry, granite appeared as a light-colored rock body with a coarse crystalline texture. The rock was mainly whitish to light grey, with visible dark and light mineral grains. This observation indicates that the granite formed through slow crystallization of magma below the surface before being exposed by later uplift, erosion, or quarrying activities.

[Image: Light-colored granite]

Figure 6: Light-colored granite observed at Sao Mai Quarry.

2.4.2 Diabase Dykes

Diabase is another intrusive igneous rock observed during the fieldtrip. Unlike granite, diabase is darker in color and usually has a finer texture. It commonly occurs as a dyke, which forms when magma intrudes into cracks or fractures of pre-existing rocks and then solidifies.

At Sao Mai Quarry, diabase appeared as a dark grey to greenish-grey rock. Its darker color and finer texture made it clearly different from the surrounding granite. The presence of diabase indicates that after the granite had already formed, new magma later intruded into fractures inside the older rock body.

2.4.3 Cross-cutting Relationship of Intrusive Igneous Rocks

One of the most important geological features observed at Sao Mai Quarry was the cross-cutting relationship between diabase and granite. In the outcrop, the darker diabase body was seen cutting across the lighter granite body. This relationship is important because it helps determine the relative age of the rocks.

According to the principle of cross-cutting relationships, a geological body that cuts through another rock must be younger than the rock it cuts. Therefore, the diabase dyke is younger than the granite. This

[Image: Dark-colored diabase]

Figure 7: Dark-colored diabase observed at Sao Mai Quarry.

means that the granite formed first, and later, diabase magma intruded into fractures within the granite and solidified.

[Image: Diabase cutting granite]

Figure 8: Dark diabase cutting through light-colored granite at Sao Mai Quarry.

This field evidence helped students understand how geologists reconstruct the sequence of geological events. Instead of only identifying rock types, students could interpret which rock formed earlier and which formed later based on their relationships in the field.

2.5 Summary of Geosphere Observations

The geosphere observations during the fieldtrip showed several important geological processes. Laterite represented intense chemical weathering under tropical conditions. Cua Lap illustrated coastal erosion, wind transportation, and sand deposition. Rhyolite demonstrated volcanic eruption processes and the formation of extrusive igneous rocks. Sao Mai Quarry showed intrusive igneous activity through granite and diabase, especially the cross-cutting relationship between the two rock types.

Overall, these observations helped students understand how rocks are formed, modified, weathered, and exposed at the Earth's surface. They also showed that geological features in the field can be used to interpret the formation history and relative timing of geological events.

3 Hydrosphere

3.1 Surface Water and Stream Processes at Suoi Tien (Fairy Stream)

Suoi Tien (Fairy Stream) represents a unique surface water system flowing through a micro-canyon terrain shaped by wind and water erosion. The shallow stream is characterized by a steady, low-velocity laminar-to-turbulent flow that constantly transports fine sand particles down to coastal estuaries.

As surface water flows along the channel, it continuously undercuts the base of the surrounding sand and clay cliffs. This lateral stream erosion, combined with rainfall runoff, causes small-scale slope instability and mass wasting along the stream margins. The interaction between surface water and weathered sand formations demonstrates active fluvial-aeolian erosion and sediment transport within a coastal stream environment.



Figure 9: Surface water flow and undercut sand cliffs observed at Suoi Tien.

3.2 Groundwater Dynamics in Coastal Sand Dune Systems (Ho Tram & Coastal Dunes)

Coastal sand dunes, such as those observed around Ho Tram and adjacent coastal stretches, play a critical role in local hydrogeology. Because loose dune sand possesses high porosity and permeability, rainwater easily infiltrates directly into the subsurface with minimal surface runoff, establishing these dune fields as effective natural groundwater recharge zones.

Rainwater penetrating through the sandy layers forms a shallow freshwater lens above denser saline water. During field observations, local groundwater wells and natural seepages along dune depressions demonstrated the availability of accessible freshwater in sandy environments. However, these shallow aquifers remain highly sensitive to over-extraction and sea-level fluctuations, which can trigger saltwater intrusion into the fresh groundwater zone.



Figure 10: Aeolian sand dune surface acting as a groundwater recharge zone at Ho Tram.

3.3 Coastal Erosion, Wave Action, and Marine Processes (Back Beach, Mui Ke Ga, & Ho Tram Estuary)

Coastal hydrological processes were investigated across multiple distinct coastal environments, including Back Beach (Vung Tau), Mui Ke Ga, and the Ho Tram river mouth / coastal inlet (Cua Ho Tram).

At Back Beach and Cua Ho Tram, continuous wave action and tidal currents drive intense coastal sediment transport. High-energy waves strike the shoreline, creating steep beach scarps and eroding sandy shorelines, while longshore currents carry loose sand along the coast to form depositional bars near river inlets.



Figure 11: Eroded sandy beach scarp and wave impact observed along the coast.

At Mui Ke Ga, where rocky headlands intersect the ocean, wave energy is concentrated directly against bedrock exposures. Mechanical wave battering and hydraulic fracturing constantly weather and erode the granitic headland structures, creating wave-cut platforms and sea cliffs. The contrasting dynamics between the sandy beaches of Ho Tram / Vung Tau and the rocky exposures of Mui Ke Ga highlight how coastlines respond differently to ocean hydrodynamics based on local geology.

3.4 Impact of Mining Pit Water Retention at Ho Da and Gia Quy Quarry

The hydrosphere also interacts closely with human engineering and resource extraction activities, as observed at Ho Da (VNU-HCM Quarry lake) and Gia Quy Quarry.

At Ho Da, past quarrying deepened the basin below the regional water table, allowing groundwater seepage and surface runoff to accumulate over time. This created a permanent artificial quarry lake. Similarly, open-pit operations at Gia Quy Quarry alter local surface runoff paths and groundwater drainage. These sites demonstrate how anthropogenic excavations interrupt natural hydrological flow paths and create standing water bodies within former mining pits.



Figure 12: Wave action against rocky granite headlands observed at Mui Ke Ga.



Figure 13: Accumulated surface and groundwater filling the former quarry pit at Ho Da.

4 Biosphere

4.1 Biological Weathering by Tree Roots

Biological weathering is the breakdown of rocks caused by living organisms such as plants, roots, microorganisms, and animals. Among these agents, plant roots are one of the most common causes of biological weathering in exposed rock areas. When seeds fall into small cracks or fractures in rocks, they can germinate if there is enough moisture and soil material. As the plant grows, its roots gradually extend deeper into the fractures.

Over time, the roots become larger and exert pressure on the surrounding rock. This pressure slowly widens the cracks and weakens the rock structure. As a result, the rock may split into smaller blocks or fragments. This process is usually slow, but it is important because it contributes to the long-term breakdown of rocks at the Earth's surface.

During the fieldtrip, vegetation was observed growing along fractures in exposed rock. The plant stems and roots followed existing cracks in the rock surface. This observation shows how biological activity can take advantage of natural weaknesses in rocks and gradually enlarge them. The presence of vegetation also indicates that weathered material and moisture had accumulated in the cracks, allowing plants to grow.

[Image: Plant growth along fractures]

Figure 14: Plant growth along rock fractures, demonstrating biological weathering caused by roots.

This example demonstrates the connection between the biosphere and geosphere. The biosphere, represented by plant growth, directly affects the geosphere by weakening and breaking down rock materials. In addition, biological weathering can support other weathering processes. Once cracks become wider, more water and air can enter the rock, increasing chemical weathering and physical breakdown. Therefore, root growth plays an important role in rock weathering and landscape development.

4.2 Mangrove Forests and Coastal Protection

Mangrove forests are coastal ecosystems that commonly develop in river mouths, estuaries, tidal flats, and sheltered coastal zones. These environments are usually affected by brackish water, which is a mixture of freshwater from rivers and saltwater from the sea. Mangrove trees are specially adapted to muddy, wet, and saline environments. They often have dense root systems that help them survive in soft sediments and tidal conditions.

At river mouths, mangrove forests act as a natural transition zone between land and sea. Their root systems trap fine sediments carried by rivers and tides. As sediments accumulate around the roots, the ground becomes more stable. This process helps reduce sediment loss and supports the formation of coastal wetlands.

[Image: Mangrove forest]

Figure 15: Mangrove forest along a water channel, showing dense vegetation in a coastal wetland environment.

Mangrove forests play an important role in coastal protection. Their dense trunks, branches, and roots reduce the energy of waves, tidal currents, and storm surges before they reach inland areas. By slowing down water movement, mangroves help limit coastal erosion and protect riverbanks and shorelines. They also reduce the direct impact of strong winds and waves during storms.

In addition, mangroves provide habitats for many organisms, including fish, crabs, mollusks, birds, and microorganisms. Their root systems create sheltered environments where young aquatic animals can live and grow. Therefore, mangrove forests are important not only for coastal protection, but also for biodiversity and local ecosystems.

[Image: Mangrove vegetation]

Figure 16: Mangrove vegetation near a river-mouth environment, illustrating its role in stabilizing sediments and protecting coastal areas.

Although mangrove forests were not the main focus of every field observation, they are highly relevant to the Earth system because they show the interaction between the biosphere, hydrosphere, and geosphere. Mangrove vegetation grows in coastal water environments, traps sediments, stabilizes shorelines, and reduces erosion. Therefore, mangrove forests are natural barriers that help protect coastal areas from waves, storms, and flooding.

5 Atmosphere

5.1 Description of Typhoon Characteristics in Vietnam

Vietnam is strongly affected by tropical cyclones because the country has a long coastline facing the East Sea. Tropical cyclones in Vietnam may appear as tropical depressions, tropical storms, severe tropical storms, typhoons, or very strong typhoons depending on their wind speed and intensity. These systems usually bring strong winds, heavy rainfall, high waves, storm surges, coastal erosion, flooding, and landslides.

5.1.1 Number of Storms per Year

On average, about 11 to 13 tropical cyclones enter the East Sea each year during the main storm season, and around 4 to 6 storms make landfall or directly affect mainland Vietnam. Some sources also report that Vietnam is affected by about 6 to 8 typhoons or tropical cyclones per year on average. The exact number changes from year to year depending on oceanic and atmospheric conditions [4, 3].

5.1.2 Time of Typhoon Occurrence

The typhoon season in Vietnam usually occurs from around June to November. However, the time of impact differs by region. Northern Vietnam is often affected earlier in the season, while Central Vietnam is commonly affected later, especially from September to November. Southern Vietnam is less frequently hit directly by typhoons, but it can still be affected by tropical depressions, heavy rainfall, strong winds, and rough seas [4, 2].

5.1.3 Magnitude and Intensity of Typhoons

Typhoons vary in intensity from weak tropical depressions to strong and very strong typhoons. A tropical depression has lower wind speed but can still bring heavy rainfall. A tropical storm or typhoon has stronger wind, higher waves, and greater destructive power. Very strong typhoons may damage houses, trees, electricity systems, boats, coastal structures, and transportation networks.

The intensity of a typhoon depends on sea surface temperature, atmospheric moisture, wind conditions, and its movement path. When a typhoon moves over warm seawater, it can gain energy and become stronger. When it moves over land, it usually weakens but may still cause heavy rainfall and flooding.

5.1.4 Affected Areas

Typhoons mainly affect coastal provinces of Vietnam, especially the northern and central coastal regions. Areas from Quang Ninh to Thanh Hoa often experience a high frequency of typhoon impacts, while Central Vietnam is highly vulnerable because of its narrow coastal plains, short rivers, steep terrain, and exposure to storms from the East Sea. Southern Vietnam is less frequently affected by direct typhoon landfall, but coastal areas can still experience heavy rain, strong winds, and sea-related hazards [4, 2].

5.1.5 Frequency of Strong Typhoons

Strong typhoons do not occur every month, but when they occur, they can cause serious damage. Their frequency varies from year to year. Some years have fewer storms, while other years have many storms and tropical depressions. Climate variability, including El Nino and La Nina conditions, can influence the number and paths of storms affecting Vietnam.

5.1.6 Consequences of Typhoons

Typhoons can cause serious impacts on both natural environments and human activities. Strong winds can damage buildings, trees, power lines, boats, and coastal infrastructure. Heavy rainfall can cause flash floods, urban flooding, river flooding, and landslides. Along the coast, high waves and storm surges can increase coastal erosion and damage beaches, dunes, and coastal settlements.

In Earth Science, typhoons are important because they show the strong interaction between the atmosphere, hydrosphere, geosphere, and biosphere. Wind and rain come from atmospheric processes, waves and storm surges affect the hydrosphere, erosion and landslides modify the geosphere, and vegetation and ecosystems can be damaged by storm impacts.

5.2 Rainfall Characteristics in Vietnam by Region

Vietnam has a monsoon-influenced climate, so rainfall is strongly seasonal and differs between regions. The distribution of rainfall is affected by monsoon winds, topography, distance from the sea, and the influence of tropical storms. In general, Vietnam has a wet season and a dry season, but the timing and intensity of rainfall vary across the country [3].

5.2.1 Rainfall in Northern Vietnam

Northern Vietnam has a humid monsoon climate with clearer seasonal changes. Summers are hot and rainy, while winters are cooler and drier. Heavy rainfall usually occurs during the summer monsoon season, especially from May to October. In many northern areas, the rainy season begins around April-May, reaches a peak around July-August, and ends around September-October or October-November, depending on location. Tropical storms and typhoons can also bring intense rainfall to northern coastal and mountainous provinces [3].

In mountainous areas of northern Vietnam, heavy rain can quickly flow down steep slopes and narrow valleys. This increases the risk of flash floods and landslides, especially where the soil and weathered rock layers are already weak.

5.2.2 Rainfall in Central Vietnam

Central Vietnam often receives very heavy rainfall, especially during the later months of the year. This region is strongly affected by typhoons, tropical depressions, northeast monsoon winds, and complex topography. Heavy rainfall is especially common from around September to December, when storms and monsoon winds strongly affect the region. For the North Central Coast, the rainy season commonly begins in August-September, peaks in October-November, and ends in November-December. For the South Central Coast, the rainy season also begins around August-September, peaks in October-November, and may continue until December [3].

Because the land is narrow and the mountains are close to the sea, rivers are usually short and steep. As a result, heavy rainfall can rapidly produce floods. Central Vietnam is one of the regions most vulnerable to storm-related disasters. Intense rain may cause river flooding, flash floods, landslides, and coastal erosion. The combination of steep terrain and heavy rainfall makes this region highly sensitive to natural hazards.

5.2.3 Rainfall in Southern Vietnam

Southern Vietnam has a tropical monsoon climate with two main seasons: the rainy season and the dry season. The rainy season usually occurs from around May to November, while the dry season lasts from around December to April. Rainfall in the south is mainly influenced by the southwest monsoon. Coastal areas in the south generally have an average annual rainfall of about 1,500 to 2,000 mm [3].

Vung Tau belongs to the southern coastal region. Therefore, rainfall in the fieldtrip area is influenced by the wet-dry seasonal pattern, coastal humidity, sea breeze, and monsoon activity. Although Southern Vietnam is less frequently hit directly by strong typhoons than Northern and Central Vietnam, heavy rainfall and coastal processes can still affect erosion, flooding, and sediment movement.

5.2.4 Comparison Among Regions

Overall, rainfall in Vietnam differs clearly by region. Northern Vietnam has a hot rainy summer and a cooler dry winter. Central Vietnam receives very intense rainfall during the storm and monsoon period, especially from September to December, making it highly vulnerable to floods and landslides. Southern Vietnam has a more distinct wet and dry season pattern, with rainfall concentrated from around May to November.

This regional difference shows that rainfall in Vietnam is controlled by both atmospheric circulation and local geography. Monsoon winds, typhoons, mountains, coastal plains, and river systems all influence how rainfall is distributed and how it affects natural hazards.

5.3 Impacts of Storms and Rainfall on Landslides and Flooding Hazards

Storms and heavy rainfall are among the main causes of natural hazards in Vietnam. When rainfall is intense or lasts for many hours or days, it can saturate soil, increase runoff, raise river water levels, and weaken slopes. These processes can lead to landslides, flash floods, river floods, and urban flooding.

5.3.1 Landslides

Landslides occur when soil, weathered rock, or fractured rock moves downslope under the influence of gravity. Heavy rainfall is one of the most important triggering factors. When rainwater infiltrates into soil and weathered rock, it increases pore water pressure and reduces the strength of the slope material. As a result, the slope becomes less stable and may collapse.

Landslides are especially common in mountainous and hilly areas where slopes are steep, rocks are fractured, and weathered materials are thick. Human activities such as road construction, deforestation, quarrying, and slope cutting can also increase landslide risk. During the typhoon season, especially from June to November, prolonged heavy rainfall can trigger many landslides at the same time, damaging roads, houses, and agricultural land [1].

5.3.2 Flooding

Flooding occurs when rainfall exceeds the ability of rivers, canals, drainage systems, or the ground surface to carry or absorb water. In low-lying areas, water can accumulate quickly and cause inundation. In urban areas, flooding may become more serious because concrete surfaces reduce infiltration and drainage systems may be overloaded.

In coastal areas, flooding can be worsened by high tides, storm surges, and strong waves. When heavy rainfall occurs at the same time as high sea level, water drains more slowly into the sea. This can increase flood depth and duration. Coastal flooding may also combine with erosion, damaging beaches, dunes, roads, and coastal settlements.

For example, during the 2020 Central Vietnam floods, several storms and tropical cyclones combined with the inter-tropical convergence zone caused widespread flooding, landslides, storm surge, and strong winds. Some central provinces recorded very large rainfall totals, with more than 2,400 mm in many

areas between 5 and 20 October 2020. This example shows how repeated storms and prolonged rainfall can create severe flooding and landslide hazards [1].

5.3.3 Relationship with Field Observations

The fieldtrip observations helped illustrate some effects of atmospheric processes on the landscape. At Cua Lap, wind and coastal processes transported sand and contributed to dune formation. Heavy rainfall and storms can intensify these processes by increasing runoff, wave energy, and sediment movement. At coastal sites such as Cua Lap and Back Beach, storms may accelerate coastal erosion and reshape the shoreline.

At quarry and hillside areas, rainfall can also increase weathering and slope instability. Water enters cracks and fractures in rocks, weakens weathered materials, and may help trigger small rockfalls or slope failures. Therefore, storms and rainfall are not only weather events, but also important factors controlling erosion, sediment transport, landslides, and flooding hazards.

6 Natural Resources

6.1 VNU-HCM Quarry and Environmental Restoration Efforts

VNU-HCM Quarry, also known as Ho Da, is a former quarry located inside the VNU-HCM area. In the past, this site was exploited for construction materials. After mining activities stopped, the abandoned quarry gradually became a lake surrounded by exposed rock slopes and vegetation. This transformation shows how a former mining site can become a landscape with educational, environmental, and recreational value.

From a geological perspective, the exposed rock surfaces around VNU-HCM Quarry provide a useful field site for observing andesite and other weathered materials. Students can study the relationship between quarrying activities, exposed rock formations, slope conditions, and surface weathering processes. The site also allows students to understand how human extraction can strongly modify the natural landscape.

[Image: VNU-HCM Quarry / Ho Da]

Figure 17: VNU-HCM Quarry, also known as Ho Da, showing the abandoned quarry lake and surrounding rock slopes.

In terms of environmental restoration, VNU-HCM Quarry is an example of partial natural recovery after mining. The quarry pit has been filled with water, while vegetation has gradually grown around the lake margins and rock slopes. These changes help reduce bare exposed surfaces and improve the visual landscape. However, the area still requires proper management because steep quarry walls, deep water, and unstable slopes may create safety risks.

Therefore, VNU-HCM Quarry demonstrates both the environmental impact of past mining and the potential for restoration after extraction stops. It is no longer only a quarry site, but also a place for geological education and environmental awareness.

[Image: Rock slope and vegetation]

Figure 18: Exposed rock slope and vegetation around the former VNU-HCM Quarry.

6.2 Basalt and Pozzolan Resources at Gia Quy Quarry

Gia Quy Quarry is an active quarry where basaltic materials and pozzolan are exploited. Basalt is a dark-colored extrusive igneous rock formed when lava cools rapidly at or near the Earth's surface. Because of

its hardness, durability, and high strength, basalt is widely used as a construction material, especially as crushed stone, aggregate, road base, and building material.

During the fieldtrip, large piles of basaltic material were observed at Gia Quy Quarry. These materials show the economic value of basalt as a natural resource. The quarry activity also demonstrates how geological materials are extracted, processed, and used to support construction and infrastructure development.

[Image: Basaltic materials at Gia Quy Quarry]

Figure 19: Basaltic materials and quarry activity observed at Gia Quy Quarry.

In addition to the quarry exposure, basalt samples were collected and observed during the fieldtrip. The basalt sample showed a dark grey to black color, which is typical of mafic volcanic rocks. Some parts of the sample displayed small holes or vesicles. These vesicles were formed when gas bubbles were trapped in the lava during rapid cooling. The presence of vesicular texture indicates that the basalt originated from volcanic activity and cooled near the Earth's surface.

[Image: Basalt samples]

Figure 20: Basalt samples collected at Gia Quy Quarry, showing dark color and vesicular texture.

The observed basalt samples help explain why basalt is valuable as a construction resource. Dense basalt is hard, strong, and durable, making it suitable for crushed stone and aggregate. Vesicular basalt, although more porous, is still useful for understanding the volcanic origin and cooling history of the rock. Therefore, the basalt at Gia Quy Quarry has both scientific and economic value.

Besides basalt, pozzolan was also observed at Gia Quy Quarry. Pozzolan is a siliceous or siliceous-aluminous material that can react with calcium hydroxide in the presence of water to form cementitious compounds. Because of this reaction, pozzolan is commonly used as a mineral additive in cement and concrete production.

Pozzolan is important in the cement industry because it can partially replace cement clinker. This helps reduce the amount of clinker required, lower production cost, and decrease environmental impact. In concrete, pozzolan can improve long-term strength, durability, and resistance to chemical attack. Therefore, the pozzolanic material at Gia Quy Quarry has both economic and environmental value.

[Image: Pozzolan material]

Figure 21: Pozzolan material observed at Gia Quy Quarry, which can be used as an additive in cement production.

Gia Quy Quarry shows the close relationship between geology and industry. The basalt and pozzolan observed at this site are not only evidence of past volcanic activity, but also important natural resources for construction and cement production.

6.3 Petroleum Service Industry in Vung Tau: Vietsovpetro Joint Venture

Vung Tau is one of the most important petroleum service centers in Vietnam. During the fieldtrip, students visited Vietsovpetro Joint Venture, where they learned about offshore petroleum exploration, drilling platforms, oil and gas production, and petroleum service activities. This visit helped students understand how Earth Science knowledge is applied in the energy industry.

At Vietsovpetro, models of offshore oil fields and drilling platforms were displayed. These models showed the arrangement of offshore structures, pipelines, platforms, and oil transportation systems. Through these exhibits, students could visualize how petroleum is extracted from offshore reservoirs and transported for processing.

The visit also included models of offshore drilling and production platforms. These platforms are complex engineering structures designed to operate in marine environments. They are used for drilling

[Image: Offshore oil fields model]

Figure 22: Model of offshore oil fields and petroleum infrastructure displayed at Vietsovpetro.

wells, extracting oil and gas, supporting workers, and connecting production systems. Geological knowledge is essential in this industry because petroleum exploration depends on understanding sedimentary basins, source rocks, reservoir rocks, traps, and subsurface structures.

[Image: Offshore platform model]

Figure 23: Offshore petroleum platform model displayed at Vietsovpetro.

In addition to models, students observed displays of rock samples, core samples, and petroleum-related materials. These exhibits showed how geological samples are used to study underground formations and evaluate oil and gas potential. Core samples are especially important because they provide direct information about rock type, porosity, permeability, and reservoir quality.

[Image: Geological and petroleum samples]

Figure 24: Geological samples and petroleum-related materials displayed at Vietsovpetro.

The Vietsovpetro visit demonstrated the practical value of Earth Science in the petroleum service industry. It showed that geology is not limited to rock observation in the field, but also plays a central role in energy exploration, offshore engineering, resource evaluation, and industrial development. Therefore, this stop helped students connect geological knowledge with real professional applications in Vung Tau.

7 Conclusion

This field trip was a truly unforgettable experience for all members of our group. It was not only a journey to observe rocks, landforms, coastal environments, and natural resources, but also a meaningful opportunity for us to connect classroom knowledge with real field observations. Through each stop of the trip, we learned how Earth Science concepts appear in the real world, from weathering, erosion, and igneous rocks to groundwater, coastal processes, mangrove forests, atmospheric hazards, and natural resource exploitation.

During the fieldtrip, our group had the opportunity to learn about different types of rocks, including intrusive igneous rocks and extrusive volcanic rocks. We observed how magma-related processes can form rocks such as granite, diabase, basalt, and rhyolite, and how these rocks are later modified by weathering, erosion, and human activities. These observations helped us understand geological processes more clearly than learning from textbooks alone.

The fieldtrip also allowed us to apply practical field skills in real situations. We used knowledge of field observation, sample collection, note-taking, photography, and geological description directly at outcrops. In particular, we had the chance to apply compass-related knowledge in the field, which helped us understand how basic field equipment can support geological observation and measurement. These experiences made Earth Science more realistic, practical, and memorable for us.

Every group member had the chance to participate in field activities such as observing outcrops, collecting samples, taking photographs, recording notes, and discussing geological features. These activities helped us realize that Earth Science is not only about theories in lectures, but also about careful observation, teamwork, patience, and curiosity in the field. The trip also allowed us to become closer as a group, as we worked together, supported one another, and shared many memorable moments throughout the journey.

We would like to express our deepest gratitude to Dr. Nguyen Thu Thuy for her dedication, enthusiasm, and tireless teaching throughout the fieldtrip. Despite the long travel schedule and challenging outdoor conditions, she continuously explained the geological features at each site with great energy and

[Image: Vietsovpetro area model]

Figure 25: Scale model of the Vietsovpetro area and facilities.

passion. Her guidance helped us better understand the meaning of each observation and inspired us to appreciate Earth Science more deeply.

We would also like to sincerely thank MSc. Le Thanh Phong and all the lecturers who organized this fieldtrip. Their careful preparation, guidance, and support created a valuable learning environment for students. Thanks to their efforts, we were able to visit important geological, coastal, and industrial sites and gain practical knowledge that would be difficult to obtain in a classroom setting alone.

In addition, our group would like to thank the supporting staff members and especially Mr. Ngo Tan Tien for assisting our group and the whole fieldtrip team during the journey. Their support helped the trip run smoothly and safely, allowing students to focus on learning, observing, and completing field tasks.

Overall, this fieldtrip will remain a precious and unforgettable memory for all members of our group. It gave us not only academic knowledge, but also practical field experience, friendship, teamwork, gratitude, and respect for nature. We believe that this fieldtrip was more than an academic activity; it was a journey of learning, appreciation, and memories that will stay with us for a long time.

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